

Assumption 3 requires that each cohort experiences the same path of treatment effects. Treatment effects need to be the same across cohorts in every relative period for homogeneity to hold, whereas for heterogeneity to occur, treatment effects only need to differ across cohorts in one relative period. The assumption of treatment effect homogeneity is therefore strong, and in Section 3.4.1, we describe how it can be violated in applied settings.

Our notion of treatment effect homogeneity does not preclude dynamic treatment effects; it only imposes that cohorts share the same path of treatment effects. The related literature sometimes formulates restrictions on the dynamics of treatment effects as another notion of treatment effect homogeneity. Athey and Imbens (2018) propose an assumption that “restricts the heterogeneity of the treatment effects over time,” which implies  $CATT_{e,\ell}$  can vary over  $e$  but not over  $\ell$ . Borusyak and Jaravel (2017) refer to one type of treatment effects heterogeneity as “only across the time horizon,” which implies  $CATT_{e,\ell}$  can vary over  $\ell$  but not over  $e$ . Callaway and Sant’Anna (2020a) allow for “arbitrary treatment effect heterogeneity” when  $CATT_{e,\ell}$  varies across cohorts and over time. Similarly, de Chaisemartin and D’Haultfœuille (2020) describe treatment effects that may be “heterogeneous across groups and over time periods.” Goodman-Bacon (2018) allow heterogeneous effects to either “vary across units but not over time” or “vary over time but not across units.” The literature has not converged on a single notion of treatment effects heterogeneity with time-varying treatment. Since researchers are interested in dynamic treatment effects when using a “dynamic” specification, we do not restrict the path of treatment effects but rather use “heterogeneity” to describe variation across cohorts only.

## 2.3 Relevance in the applied literature

To gauge the empirical relevance of our results, we survey the estimation methods used by the twelve papers collected by Roth (2019) from three leading economics journals that contain the phrase “event study” in their main text.<sup>1</sup> From this sample of applied papers, we learn what specifications empirical researchers are actually using when estimating two-way fixed effects regressions. Four papers in this sample consider the simple setting where units either receive their first treatment at the same time or never receive the treatment. The other eight papers in this sample consider the more complex setting where treated units receive their

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<sup>1</sup>We follow the selection criteria in Roth (2019): the original sample consists of 70 total papers, but is further constrained to these twelve papers with publicly available data and code. The data and code are used to determine exactly the specification estimated in these papers.

treatment at various times, and there may or may not be never treated units. This observation suggests an *event study* in the applied literature nests two popular research designs: *difference-in-differences* design, but also the design where units receive their first treatments at various times, which is our focus.<sup>2</sup>

We summarize the main specifications in this sample of twelve applied papers in Table 1.<sup>3</sup> The columns of Table 1 collect key properties of these specifications. In Section 3.1, we introduce a general class of specifications that encompasses all of these specification. The estimates for relative period coefficients  $\mu_\ell$  from all these papers therefore fall under our analysis in the next section.

These papers demonstrated that event studies are used to address a broad range of research questions. As an example of this literature, Bailey and Goodman-Bacon (2015) use the rollout of the first Community Health Centers (CHCs) to study the longer-term health effects of increasing access to primary care. As another example, Tewari (2014) uses variation in the timing of deregulation across states to estimate the impact of financial development on homeownership.

### 3 Estimators from linear two-way fixed effects regression

We consider a two-way fixed effects (FE) regression of the following form, estimated on a panel of  $i = 1, \dots, N$  units for  $t = 0, 1, \dots, T$  calendar time periods:

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{g \in \mathcal{G}} \mu_g \mathbf{1}\{t - E_i \in g\} + v_{i,t} \quad (4)$$

Here  $Y_{i,t}$  is the outcome of interest for unit  $i$  at time  $t$ ,  $E_i$  is the time for unit  $i$  to initially receive a binary absorbing treatment, and  $\alpha_i$  and  $\lambda_t$  are the unit and time fixed effects. The set  $\mathcal{G}$  collects disjoint sets  $g$  of *relative periods*  $\ell \in [-T, T]$ . We allow some relative periods to be excluded from the specification and denote the excluded set with  $g^{excl} = \{\ell : \ell \notin \bigcup_{g \in \mathcal{G}} g\}$ . We denote by  $\mu_g$  the relative period coefficients from regression (4), i.e. the population regression coefficients. Their corresponding OLS estimators are denoted by  $\hat{\mu}_g$  respectively.

We are interested in the properties of  $\mu_g$  when there are variations in the initial treatment timing, and

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<sup>2</sup>This setup is the same as the *staggered adoption* design proposed by Athey and Imbens (2018), but we keep the term *event study* because it is common in the applied literature.

<sup>3</sup>We focus on the first specification underlying the event study estimates in each paper, which we view as a reasonable proxy for the main specification in the paper.

there may or may not be never-treated units. Below in Section 3.1 we illustrate how the choice of  $\mathcal{G}$  coincides with a large number of specifications encountered in practice such as the “fully dynamic” specification. We next decompose  $\mu_g$  in terms of  $CATT_{e,\ell}$  when various combinations of the three identifying assumptions fail. For Propositions 1-3, we state the results in terms of the general specification (4). To specialize these results to the “fully dynamic” specification (7), we note the corresponding decomposition would replace bins with  $g = \{\ell\}$  for each relative period  $\ell$  included in the specification and  $g^{excl}$  would contain all excluded relative periods. The decomposition remains unchanged though the summation over  $\ell \in g$  simplifies since each  $g$  is a singleton. For Proposition 4, the decomposition further simplifies for the “fully dynamic” specification as discussed below.

Researchers may assume that  $\mu_g$  can be interpreted as a convex average of  $CATT_{e,\ell}$  for periods  $\ell \in g$  from its corresponding set  $g$ ; they may further assume the underlying weights have policy-relevant interpretation, e.g. weights depending on proportions of cohorts. For example, Bailey and Goodman-Bacon (2015) interpret them as “intention-to-treat effects” of the treatment in a given relative year. However, our results show that  $\mu_g$  may not represent the parameter of interest without strong assumptions such as treatment effect homogeneity. Section 3.6 provides intuition for these negative results, and demonstrate the weights are actually non-linear functions of proportions of cohorts. Section 3.7 illustrates how treatment effects heterogeneity invalidates the pretrends test for a simple three-period setting.

### 3.1 Common specifications

Common specifications can be broadly categorized as either “static” or “dynamic”. Static specifications estimate a single treatment effect that is time invariant. In contrast, dynamic specifications allow for non-parametric changes in the treatment effects over time. Within dynamic specifications, researchers also need to address issues of multi-collinearity, and may bin or trim distant relative periods. All of these choices can be written as instances of (4) with the correct specification of  $\mathcal{G}$ , meaning that our results are applicable for a wide range of specifications employed in the empirical literature.

To clarify how to specify  $\mathcal{G}$  in regression (4), we define  $D_{i,t}^\ell := \mathbf{1}\{t - E_i = \ell\}$  to be an indicator for unit  $i$  being  $\ell$  periods away from initial treatment at calendar time  $t$ . For never-treated units  $E_i = \infty$ , we set  $D_{i,t}^\ell = 0$

for all  $\ell$  and all  $t$ . We can represent the relative period bin indicator as

$$\mathbf{1}\{t - E_i \in g\} = \sum_{\ell \in g} \mathbf{1}\{t - E_i = \ell\} = \sum_{\ell \in g} D_{i,t}^\ell. \quad (5)$$

**Static specification.** For a “static” specification  $\mathcal{G}$  contains a single element equal to  $g = [0, T]$ . The indicator  $\mathbf{1}\{t - E_i \in g\}$  is equivalent to an indicator for whether unit  $i$  has received its initial treatment by  $t$ :  $\mathbf{1}\{E_i \leq t\}$ . The “static” specification thus takes the following form

$$Y_{i,t} = \alpha_i + \lambda_t + \mu_g \sum_{\ell \geq 0} D_{i,t}^\ell + v_{i,t} \quad (6)$$

and the corresponding set of excluded relative periods is  $g^{excl} = [-T, -1]$ .

**Dynamic specification.** “Dynamic” specifications encompass any specifications of  $\mathcal{G}$  where  $\mathcal{G}$  contains more than one element, thus treatment effects are allowed to vary over time non-parametrically. In its most flexible form, a “fully dynamic” specification takes the following form

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{\ell=-K}^{-2} \mu_\ell D_{i,t}^\ell + \sum_{\ell=0}^L \mu_\ell D_{i,t}^\ell + v_{i,t} \quad (7)$$

and the corresponding set of excluded relative periods is  $g^{excl} = \{-T, \dots, -K-1, -1, L+1, \dots, T\}$ .

Excluding some relative periods from the “fully dynamic” specification is necessary to avoid multi-collinearity, either among the relative period indicators  $D_{i,t}^\ell$ , or with the unit and time fixed effects. For example, when there are no never-treated units i.e.  $\infty \notin \text{supp}(E_i)$  but with a panel balanced in calendar time, we need to exclude at least two relative period indicators in  $\mathcal{G}$ . These collinearities are discussed by Borusyak and Jaravel (2017): one multi-collinearity comes from the relative period indicators summing to one for every unit  $\sum_{\ell \in [-T, T]} D_{i,t}^\ell = 1$ , and the other multi-collinearity comes from the linear relationship between two-way fixed effects and the relative period indicators, namely  $t - E_i = \ell$ .

Excluding relative periods close to the initial treatment is common in practice. Normalizing relative to the period prior to treatment is the most common - six out of the eight papers we survey do so, as reflected in the above specification where we drop  $D_{i,t}^{-1}$ . The remaining two papers exclude  $D_{i,t}^0$ .

Excluding distant relative periods is however less common (only one of the eight papers we survey does so). Instead researchers “bin” or “trim” distant relative periods. For “binning”, researchers bin distant